Canotia Torr. ex A. Gray: History, Geography, and Etymology

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In 2005, when choosing a name for this journal, which specializes in papers on Arizona botany and mycology, the editor, L. R. Landrum, selected the generic name *Canotia.* This single-word title is concise and comes from a genus emblematic of the arid regions of Arizona. *Canotia holacantha* is a spiny small tree of the Sonoran and Mohave Deserts, with the core of its distribution found in central and western Arizona, usually at elevations between 700 and 1500 m. It is often called *crucifixion thorn* or *corona de cristo* (names used for *Castela emoryi* and *Koeberlinia spinosa* as well) or *Mohave thorn.* "*Qasi*" is the Hualapai name (Carrie Cannon, pers. comm.). Our curiosity about the origin of the name *Canotia* led us to write this paper. We here discuss the history and geography of *Canotia* and present our speculations on the etymology of the name. Scientific names used here are listed with their authors in Table 1 at the end of this paper.

History. The genus *Canotia* consists of two species: the original *C. holacantha*, type of the genus, found mainly in Arizona and the focus of this paper; and *C. wendtii,* a rare species of the Chihuahuan Desert. In the mid-1800s, after the war with Mexico (1846–1848), the United States government sponsored expeditions to explore the newly acquired territory, especially in search of potential routes for a railroad from the eastern United States to the west coast. On one of those expeditions Lt. A. W. Whipple (1817–1863) led surveyors and explorers approximately along the 35th parallel of latitude from Fort Smith, Arkansas, to Los Angeles, California (Gordon 1988). The expedition passed through Albuquerque, New Mexico, and in Arizona, near the future site of Flagstaff and along the Big Sandy and Bill Williams rivers near Wikieup. Dr. J. M. Bigelow (1804–1878), serving as "surgeon and botanist," was one of five scientists attached to the expedition, and the oldest at about 50 years old. Lt. Whipple (about 36 years old) was a topographical engineer in charge of the expedition of over 100 men (U. S. citizens, Native Americans, Mexicans, and two European scientists), while passing through Arizona. At the time of Bigelow's discovery of *Canotia holacantha*, the mules pulling the wagons were very weak, sometimes dying, wagons were being abandoned, the men were on half rations, some unvaccinated men had contracted smallpox, and the Colorado River had still not been reached. Despite this, Bigelow enthusiastically collected plants (Gordon 1988, pp. 156–182). He found an unusual leafless shrub and reported that "the Mexicans call it *canotia*" (Torrey 1857), the first specimen being collected on Jan 27, 1854. The next day, Whipple recorded in his diary that "Canotias are mingled with cedars upon the dry arroyos" (Foreman 1968, p. 204). John Torrey (1796–1873) described new species from the expedition (Torrey

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1857), and his type specimens are at the New York Botanical Garden (NY). Two specimens of *Canotia holacantha* at NY are labeled as type material. We are purposely not selecting a lectotype (which may have already been done) but we include an image of the superior specimen with a printed, official label (Fig. 1).

The second specimen has two handwritten labels attached and a sketch of a flower (Fig. 2). The lower label has the date Jan 27, 1854, the same date as the specimen in Fig. 1. The handwriting is barely legible but appears to say, "called by [Sonoran] Mexicans Canotia…near White Cliff Creek…generally 8–10 (–20 ft) high". This may be same White Cliff Creek listed in *Arizona Place Names* (Barnes 1988), which has also been called Cottonwood Creek and Knight's Creek. The lower label also mentions a figure in "Emory's Report". Emory's *Report on the United States and Mexican Boundary* includes a contribution on botany by Torrey (1858), published a year after Torrey's paper describing *Canotia* (Torrey 1857). The 1858 manuscript would certainly be available to Torrey, and probably available to whoever wrote the note. In Emory's Report there is a figure of *Holacantha emoryi* (= *Castela emoryi*), which can easily be confused with *Canotia*. The upper handwritten label has the date Feb 15, 1854, and appears to say "Canotia, Mex name…gravelly hills near Bill Williams Fork."

John Torrey described Bigelow's plant as *Canotia holacantha* but had doubts as to what family it belonged; he chose Hypericaceae (Torrey 1857). Unfortunately, he wrote that his name was "provisional" and according to modern rules of nomenclature that causes the name to be illegitimate. A few years later Asa Gray (1810–1888) reported the species name again, referring to Torrey's work, without "provisional" in his text and assigned it to Saxifragaceae without explaining why (Gray 1861). That made the name legitimate, and it is now cited as *Canotia holacantha* Torr. ex A. Gray. So, both authors get credit. To what family it belongs has remained controversial. In 1865, J. D. Hooker (1817-1911) chose Rosaceae with some doubt, based mainly on fruit characters. Gray seems to have sent a flowering specimen collected by C. C. Parry (1823–1890) to a French botanist, H. E. Baillon (1827–1895), who suggested the family Celastraceae as a home for *Canotia* (Baillon 1873). Gray (1877) proposed Rutaceae; Britton (1908) said the species was so distinct that it deserved its own family, Canotiaceae; Barnhart (1910) assigned *Canotia* to Koeberliniaceae. Molecular studies (Simmons et al. 2001) based on the phytochrome B gene sequence and morphology support the placement of *Canotia* in Celastraceae and show that it is related to *Acanthothamnus* of Mexico.

Until recently, it was thought that there were three species of Celastraceae in Arizona *Canotia holacantha*, *Mortonia scabrella*, and *Paxistima myrsinites* (Brasher 1998), all woody plants. The molecular studies (Simmons et al. 2001) have shown that *Parnassia palustris*, an herb, is also a member.

Whipple had risen to the rank of Brigadier General by the time of the Civil War and was mortally wounded at the battle of Chancellorsville in 1863. Fort Whipple, near Prescott, Arizona, was named in his honor in 1864 (Wikipedia 2026). The name *Canotia* has been commemorated as a street name, "Canotia Place," in Cave Creek, Arizona; and the USS *Canotia* named after the tree was one of 35 wooden-hulled, relatively small, *Ailanthus*-class ships built during World War II (Fig. 3). It served mainly in defensive operations in the South Pacific, including Iwo Jima in 1945. The ship received a Battle Star award for combat service. Perhaps because *Ailanthus*-class ships were wooden they were frequently named for woody plants, e.g., *Catclaw*, *Chinaberry*, *Shellbark*, and *Viburnum* (Wikipedia 2026).

Geography. Although *Canotia holacantha* is most commonly found in central and western Arizona, it has been collected as far north as the Grand Canyon and as far south as Sonora, Mexico (SEINet 2026). A Pleistocene fossil record from an 11,100 yr B.P. packrat midden from Picacho Peak, Pinal County, Arizona (Van Devender et al. 1991) documents southeastward expansion of this species in the late Wisconsin period in pinyon-juniper woodlands. Recently, isolated populations of *C. holacantha* were discovered in the Waterman Mountains in Pima County (*Van Devender 2002-1152*, ARIZ) and 9 km west of Oracle in Pinal County (*Matthew B. Johnson 22-001*, ARIZ), 32 km south and 46 km east of Picacho Peak, respectively. In Nevada, it is only known from near Laughlin just west of the Arizona border. Records for California and Baja California were based on misidentifications of *Castela emoryi* or *Koeberlinia spinosa*. In Sonora, it is known from the Ímuris, Magdalena, and Tubutama areas. Wilder et al. (2008) reported it on Isla Tiburón in the Gulf of California. Recently, it was reported from Sonoran desert-scrub on a limestone slope in the Sierra El Viejo (Municipality of Pitiquito, 3.9 km [by air] SW of La Unión, 51.8 km [by air] E of Puerto Lobos, 30.22299°N 112.31506°W, 428 m (*J. Ortiz-V.*, iNaturalist).

Using the SEINet database with the Nearby program developed by Daryl Lafferty and our own observations, we see that plants that commonly grow with *Canotia* in Arizona are *Ephedra* spp., *Eriogonum wrightii, E. fasciculatum, Juniperus arizonica, Keckiella antirrhinoides, Krameria erecta, Larrea tridentata, Quercus turbinella, Bothriochloa barbinodis, Condaliopsis divaricata* (*= Ziziphus obtusifolia*), *Berberis haematocarpa, Rhus aromatica, Vachellia constricta* (*=Acacia constricta*), and *Simmondsia chinensis.* This seems to be a mixture of upland Sonoran desert species and others more typical of chaparral and pinyon-juniper.

Etymology. Several Mexican workers were attached to the Whipple expedition (Gordon 1988), and Bigelow would have learned the name *canotia* from them. We think the key to understanding the etymology of *Canotia* is its resemblance to *Ephedra* (joint-fir, Mormon tea, Fig. 5), a group of primitive gymnosperm shrubs that have jointed stems like those of *Canotia* (Figs. 1 & 4). *Ephedra aspera* is widespread in the southwestern United States from California to central Texas and is common along the Whipple expedition route in Mohave County, Arizona. It also grows in the Chihuahuan Desert to central Mexico and on the Baja California Peninsula. Today, common Spanish names for *Ephedra* in southwestern United States and Mexico are *cañutilla* or *cañatilla.* The masculine version, *cañutillo*, has been used as the name of towns in Texas and Durango, Mexico. Villanueva-Almanza (2010) reports these names and a few others for species of *Ephedra* in Mexico. As spoken words, the accent is on the "i" in both, while "ll" is pronounced like an English "y." The difference between "*ía*" and "*illa*" is small and would probably sound the same to an English-speaking person. Perhaps the Mexicans who provided the name to Bigelow thought that *Canotia holacantha* was an *Ephedra* that they knew as *cañutilla* or *cañatilla*, and to Bigelow the Mexican name sounded like *canotia*.

The stems of *Canotia* and *cañutilla (Ephedra*) both resemble "*caña*" (from the Latin *canna* - a cylindrical, jointed stem with hollow internodes (Lexipedia, 1999) like giant cane (*Arundo donax*) in the Old World and giant reed (*Phragmites australis*) in the New World. The ending "*illa*" [*-illo, -ita,* and *-ito*] implies small size; e,g., male - *árbol*: *arbolillo*, female - *mesa*: *mesilla*]. *Cañutilla* or *cañatilla* means small or perhaps narrow cane.

We realize this is speculation. We still have not found evidence of a name like *canotia* or c*añutilla* being used as a vernacular name for *Canotia holacantha* in modern times. But we

believe the morphological resemblance between *Ephedra* and *Canotia* has been a factor in the origin of the latter's scientific name. If our speculations are correct, the pronunciation of *Canotia* has changed. Bigelow would have said "ka-noh-TEE-ah" whereas today in Arizona it is generally called "kuh-NOH-shuh."

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Figure 1. Type specimen of *Canotia holacantha*. (Photo from JSTOR Global Plants).



Figure 2. Two handwritten labels from original material of *Canotia holacantha*. Sketch on right of old flower or young fruit. (Photo from JSTOR Global Plants).

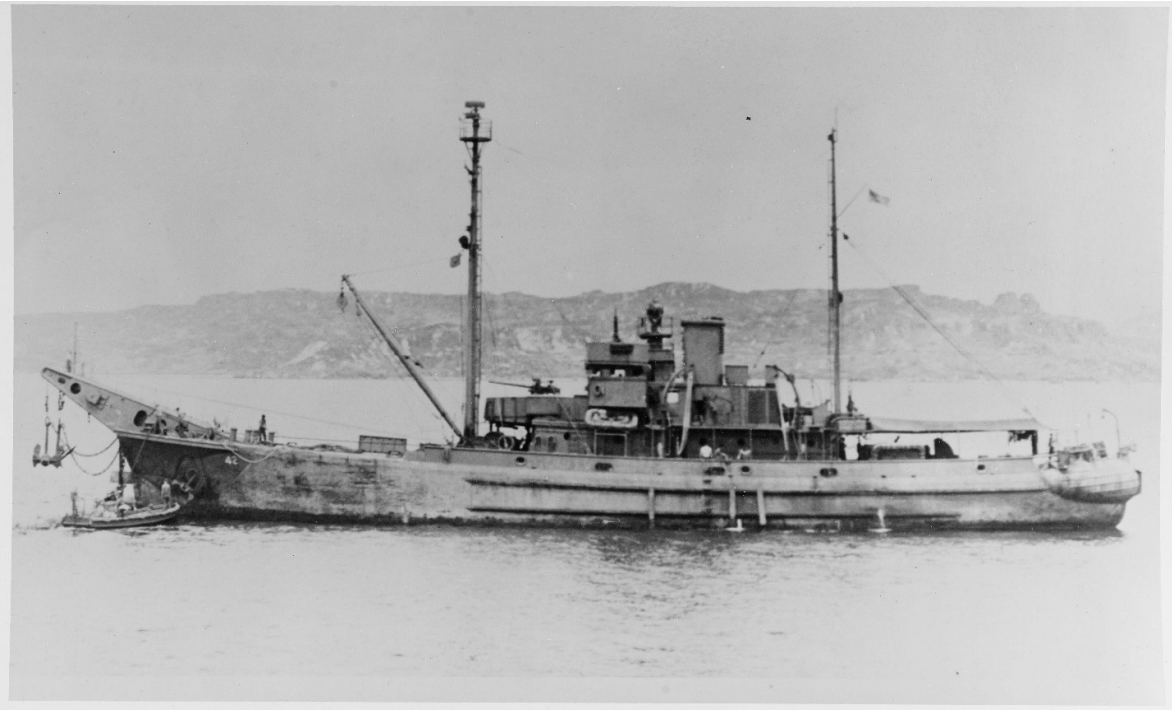


Figure 3. *Ailanthus*-class ship USS *Cliffrose* in 1945. (Photo from Wikipedia).

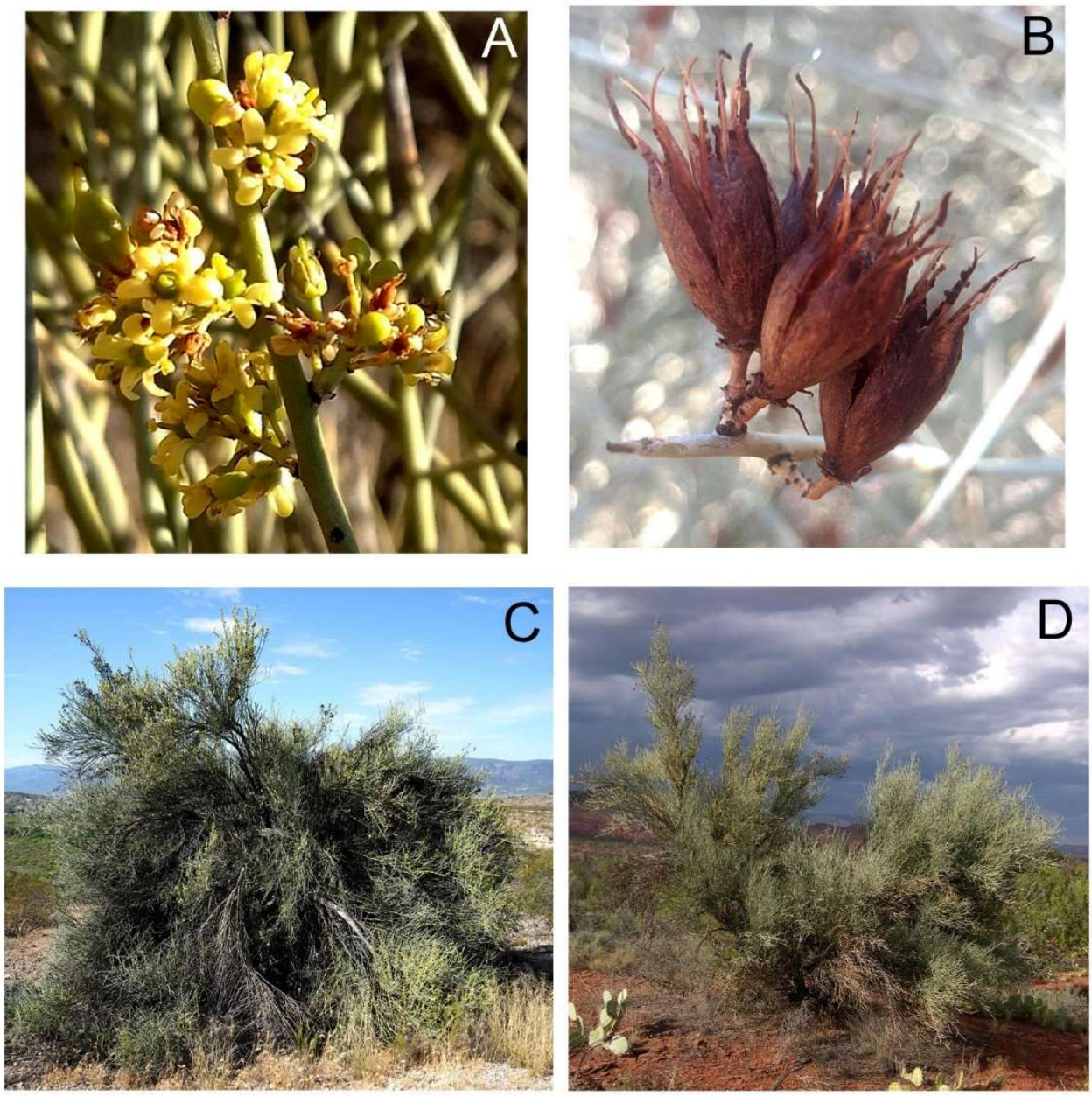


Figure 4. *Canotia holacantha.* **A**. Stems with open flowers, Sierra el Viejo, Sonora. **B**. Fruits after opening, Desert Botanical Garden, Phoenix, Arizona. **C**. Plant growing near Camp Verde, Arizona. **D**. Plant growing near Sedona, Arizona. (Photo credits: **A**, J. Ortiz-V.; **B**, L. R. Landrum; **C**, R. A. Villa; **D**, M. Licher).



Figure 5. *Ephedra aspera.* **A**. Herbarium sheet from between Jacumba and Incopa, California. **B**. Mature seed cones, Tinaja Altas, Arizona. **C**. Open staminate cones, Usery Mountain Regional Park, Arizona. (Photo credits: **A**, ASU herbarium; **B**, S. D. Carnahan; **C**, E. Makings).

Table 1. List of scientific names with authors used in this paper

Acacia constricta Benth.
Acanthothamnus Brandegee
Arundo donax L.
Berberis haematocarpa Wooton
Bothriochloa barbinodis (Lag.) Herter
Condaliopsis divaricata (A. Nelson) G. L. Nesom
Canotia Torr. ex A. Gray
Canotia holacantha Torr. ex A. Gray
C. wendtii M. C. Johnst.
Castela emoryi (A. Gray) Moran & Felger
Ephedra aspera Engelm. ex S. Watson
Eriogonum fasciculatum Benth.
E. wrightii Torr. ex Benth.
Holacantha emoryi A. Gray
Juniperus arizonica (R. P. Adams) R. P. Adams
Keckiella antirrhinoides (Benth ex A. DC.) Straw
Koeberlinia spinosa Zucc.
Krameria erecta Willd.
Larrea tridentata (Sessé & Moc. ex DC.) Coville
Mortonia scabrella A. Gray
Parnassia palustris L.
Paxistima myrsinites (Pursh.) Raf.
Phragmites australis (Cav.) Trin. ex Steud.
Quercus turbinella Greene
Rhus aromatica Aiton
Simmondsia chinensis (Link) Schneid.
Vachellia constricta (Benth.) Seigler & Ebinger
Ziziphus obtusifolia (Hook. ex Torr. & A. Gray) A. Gray